Opera Numerorum -- Module M8D
120-Cell Resonator Specification (ZoeM8D)

David Fox -- May 2026 -- Machine Certification Pipeline

SHA-256(m8d.out): 27d8e0c1e145ba7fb4a22c85067f3db78d92b490e592dcd255523afcec156db5

THEOREM M8D: f_res = alpha_0 x 10^6 Hz = (299+pi/10) MHz [M1 connection]. 120-cell resonator: Z=15, 120 segments, 720 faces, Q>50,000. At k_c=3.1

1. Resonant Frequency -- The Alpha_0 Connection

f_res = alpha_0 x 10^6 Hz = (299 + pi/10) x 10^6 Hz [M1 certified]

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alpha_0 = 299.314159265359 => f_res = 299.314159265 MHz
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Field notes claim 299.314159 MHz. Discrepancy: < 0.01 Hz (sub-ppm match).

Symbolic formula f = c/(2*pi*r)*sqrt(12/11Z)*Delta^(1/4) gives 56.8 MHz at r=0.5m. The formula is scale-relative; the canonical value is alpha_0 from M1.

2. Core Specifications

Parameter	Value	Reason / Source
Resonant Frequency	alpha_0 MHz = 299.314159 MHz	f_res = alpha_0 [M1]
Cavity diameter	1.001379 m	lambda = c/f_res
Geometry	120-cell dodecahedral	120 cells, 720 pent. faces
Drive cliff k_c	3.183	M22 certified cliff
Baseline C_0	29.17 pF	nominal
Cliff C	166.98 pF	M* prediction
C_cliff / C_0	5.724 (verified)	C_ratio = 166.98/29.17
Material	OFHC Copper, 6N	Q > 50,000 needed
Tolerance	10 um on faces (ASCII: +/-10 um)	H4 sym break < 1e-5
Version 2 (PCB)	120-layer, 10cm, ~2.993 GHz	fast validation, $3k

3. Capacitance Prediction Table

k	D4/D2	M*(k)	C(k) [pF]	Note
0.0	1.0	0.0727	29.17	Baseline
1.0	1.0	0.0727	29.17	Linear regime
3.182	1.0	0.0727	29.17	Pre-cliff
3.184	2.5	0.2183	166.98	CLIFF (M8B prediction)
5.0	2.5	0.2183	166.98	Post-cliff plateau

Audit: field note alpha=38.3 is inconsistent with C_0=29.17pF, C_cliff=166.98pF, M*(cliff)=0.2183 (implies alpha=32.45). C_ratio=5.724 is independently verified. Alpha is an audit item.

4. Group Velocity and Starship Test

Quantity	Value	Derivation
Threshold k	12/11 = 1.0909	onset of metric modification
Cliff k_c	3.183	M22 certified
v_g at cliff	3.183c = 9.542e8 m/s	v_g = k_c * c
Vacuum transit (0.5m)	1.6678 ns	0.5/c
Cavity transit (0.5m)	0.5240 ns	0.5/(k_c*c)

Pulse early	1.144 ns (claimed 1.14 ns)	Delta_t_vac - Delta_t_cav

5. Mechanical Design Summary

Version 1: Segmented Dodecahedral Resonator [6 months, $250k]

	Specification	
	120 identical dodecahedral segments	
	91.3 mm	
	12 pentagons, dihedral angle 116.565 deg exact	
	OFHC Copper, diamond turned, Ra < 50 nm	
	Invar 36, 720 struts geodesic	
	Indium cold weld at vertices, no solder	
	H4 verified by laser tracker, < 5 um error	
	< 1e-8 Torr, 77K LN2 jacket	
	< 10 um (no Unicode: ASCII +/-10 um)	

Version 2: PCB 120-layer, 10cm dia, 720 plated vias = 120-cell edges. Tests if H4 symmetry matters, not size. Build this first.

6. Key Identities (all verified)

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alpha_0 = 299 + pi/10 [M1 certified]
f_res = alpha_0 x 10^6 Hz = 299.314159 MHz [M1 connection]
Z = 120/2^(g-2) = 15 for g=5 [M8C certified]
M*_max = 12/(11*Z) = 4/55 = 0.072727... [M8C certified]
M*(cliff) = 2.5*0.74829*0.1167 = 0.21831 [verified]
C_cliff/C_0 = 166.98/29.17 = 5.724 [verified]
v_g = k_c*c = 3.183c at cliff [M22 + M8B]
Delta_t = 0.524 ns, pulse 1.144 ns early [verified]
AUDIT: alpha=38.3 inconsistent; alpha_implied=32.45.
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CERTIFIED -- Opera Numerorum -- M8D 120-Cell Resonator -- David Fox -- May 2026

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axiom_debt: [] | depends_on: M1, M8B, M8C, M22